TRACK 1: VARIANT PREDICTION

Mosaic Variegated Aneuploidy (MVA) Hackathon 2026 • Methods & Model Report

Team: Los Tanos
Model: Model 1 (Automated Pipeline)
Genome: GRCh38 / hg38

Abstract & Method Overview: We developed an automated pipeline to prioritize causal genetic variants from whole-genome sequencing (WGS) data. It combines sequencing-quality filters and allele-aware subtraction of common variants with local snpEff annotation, multi-source pathogenicity scores, ontology-aware (Resnik) phenotype similarity and compound-heterozygous pairing. Starting from 5,012,204 variants, the pipeline ranked first a pair of rare heterozygous variants in **BUB1B**, the MVA type 1 gene, consistent with a compound-heterozygous configuration (phase unconfirmed), with an EPCR of **0.9379** against 0.5627 for the next candidate.

5.01M	591k	905	0.9379
RAW INPUT VARIANTS	RARE, HIGH-QC (88.2% REMOVED)	PROTEIN-ALTERING (HIGH/MODERATE)	TOP EPCR (BUB1B PAIR)

1. FIVE-STAGE COMPUTATIONAL ARCHITECTURE

The steps run sequentially; each consumes the previous step's output, and the submission file is produced without manual editing:

STEP 1
Phenotype Extraction
8 HPO terms parsed from the clinical document; rhabdomyosarcoma is the strongest MVA anchor.

STEP 2
VCF Pre-Filtering
GQ ≥ 20, DP ≥ 10, PASS, AB 0.30–0.70 for hets, allele-aware dbSNP Common subtraction.

STEP 3
Annotation
Local snpEff GRCh38.99 (canonical transcripts), MyVariant.info and AlphaGenome; HIGH/MODERATE kept.

STEP 4
Phenotype Matching
Resnik similarity to the patient's HPO profile; BUB1B ranked #2 of all candidate genes (0.632).

STEP 5
Ranking & EPCR
gnomAD AF ≤ 1%, compound-het pairing and a weighted geometric-mean EPCR with tail decay.

2. METHODOLOGY DETAILS & FILTERING LOGIC

- **Phenotype (Step 1):** the 8 HPO terms come pre-tabulated in the clinical document. The strongest anchor is ***HP:0002859*** (Rhabdomyosarcoma), an established cancer predisposition in MVA. ***HP:0200067*** (Recurrent spontaneous abortion) is suggestive of chromosomal instability but has no direct HPO annotation to any MVA gene, so it carries little weight in the gene scores.
- **Quality control & common-variant subtraction (Step 2):** calls were filtered on intrinsic metrics (FILTER PASS or missing, ***GQ ≥ 20***, ***DP ≥ 10***, ***0.30 ≤ AB ≤ 0.70*** for heterozygous calls), which removed 11.7% of calls. Allele-aware subtraction against dbSNP Common (b151, ***MAF ≥ 0.01***) then left **591,339** of 5,012,204 variants (88.2% removed in total).
- **Annotation (Step 3):** a snpEff GRCh38.99 database built locally from Ensembl release 99 (one canonical transcript per variant), MyVariant.info (hg38: gnomAD, CADD, REVEL, ClinVar, OMIM) and AlphaGenome Atlas (AVI_PHRED, SNVs only). **905** HIGH/MODERATE variants were kept. dbSNP Common (2018) misses most common coding variants: 712 of the 905 have gnomAD AF > 1%, so gnomAD is the effective frequency filter, applied in Step 5.
- **Phenotype matching (Step 4):** each gene carrying a qualifying variant was scored against the patient's HPO profile with Resnik semantic similarity (HPO release 2026-09-01). *BUB1B* scored 0.632 (3 exact term matches), #2 of all candidate genes, and is the only Tier 1 MVA panel gene with rare HIGH/MODERATE variants.
- **EPCR (Step 5):** four components in [0, 1] (phenotype 0.35, pathogenicity 0.30, inheritance 0.20, frequency 0.15) are combined by a weighted *geometric* mean, so weakness on any axis pulls the score down. MVA panel genes receive a 1.2× bonus; two rare heterozygous variants in one gene are paired into one row. The top score is capped at 0.95 and each following row is at most 0.6× the previous one, which separates a single confident answer from the tail, as F-max rewards.

Submission Nature & Automation

The submission file is the **unedited, automated output** of the pipeline: ranking, pairing and EPCR were computed programmatically. After generation we reviewed the top finding and the filters for plausibility (Section 4); this review did not change any score or rank.

3. PRIMARY VARIANT FINDINGS & BIOLOGICAL CONCORDANCE

The pipeline ranked first two rare heterozygous variants in **BUB1B**, paired in one row as a candidate compound-heterozygous configuration. The row has an EPCR of **0.9379**, well separated from the runner-up (*LZTR1*, single heterozygous stop-gain, EPCR = 0.5627). Phase cannot be determined from a single-sample VCF.

Attribute	Variant 1	Variant 2
Gene & OMIM	*BUB1B* (OMIM *602860; Mosaic Variegated Aneuploidy syndrome 1, #257300)	
Coordinates (GRCh38)	chr15:40209701 (ref_1: T, alt_1: G)	chr15:40220612 (ref_2: T, alt_2: G)
Molecular consequence	Exon 17: stop-gain (TTA → TGA, p.Leu751*)	Exon 23: missense (AAT → AAG, p.Asn1016Lys)
Impact tier	HIGH (premature stop; predicted nonsense-mediated decay)	MODERATE (C-terminal pseudokinase domain)
Frequency & zygosity	gnomAD AF ≈ 3×10^{-5}; het, AB 0.54	Absent from gnomAD; het, AB 0.46
Pathogenicity evidence	CADD 36.0 | AVI_PHRED 33.8 | ClinVar: Pathogenic/Likely pathogenic (1 submitter)	CADD 24.5 | REVEL 0.47 | AVI_PHRED 25.6 (AlphaMissense-driven) | not in ClinVar
Finding classification	**Candidate compound-heterozygous pair (phase unconfirmed) — EPCR 0.9379 (primary finding)**	

Protein numbering follows the Ensembl release-99 canonical transcript used by snpEff (1,064 residues). In UniProt O60566 (1,050 residues) the same residues are Leu737 and Asn1002.

Biological Rationale & Phenotype Correlation

BUB1B encodes BubR1, a pseudokinase of the spindle assembly checkpoint (SAC) that inhibits the APC/C to prevent premature chromosome segregation. Biallelic *BUB1B* variants that reduce BubR1 cause **Mosaic Variegated Aneuploidy syndrome 1**: mosaic aneuploidies, growth deficiency, microcephaly and a high risk of childhood embryonal tumours, notably **rhabdomyosarcoma** and Wilms tumour. The child's rhabdomyosarcoma (***HP:0002859***) therefore matches the gene directly. Recurrent pregnancy loss (***HP:0200067***) is consistent with chromosomal instability but is not, by itself, specific to MVA.

4. POST-GENERATION REVIEW & DESIGN CHOICES

- **Quality audit:** a spot-check of 300 common SNVs showed the dbSNP Common file misses many common coding variants; hence the downstream gnomAD AF ≤ 1% filter.
- **Expected effect of the pair:** the stop-gain allele is expected to be lost to nonsense-mediated decay (an escaped product would lack the pseudokinase domain). The missense change lies inside that domain and is probably hypomorphic, with no functional assay yet. *In trans*, the pair would strongly reduce BubR1, the established mechanism of MVA1.
- **Scope:** non-coding variants and incidental findings were not reported, to limit false discoveries; all ten rows are *primary*. Single heterozygous variants in genes with weak phenotype similarity (e.g. *SGO1*, *MAD1L1*) are penalized.

5. GENERATIVE AI SPECIFICATION & DATA GOVERNANCE

Generative AI (provider, plan, data handling): AI assistants were used for coding, literature search and writing. Google Gemini Pro — **[plan; training opt-out setting]**. Anthropic Claude Opus — **[plan; training opt-out setting]**. No outputs were rated or flagged for feedback.

Public data: Ensembl release 99, dbSNP b151 (Common, 2018), gnomAD (via MyVariant.info), HPO (2026-09-01), MyVariant.info, AlphaGenome Atlas, CADD, REVEL, ClinVar, OMIM. **Proprietary data:** none.

6. PIPELINE EVALUATION, STRENGTHS & LIMITATIONS

Key Strengths

• Scripted end to end; code and parameters in the public repository.
• Compound-heterozygous pairing within a gene, scored as one row.
• Score separation designed for F-max (one confident row, decaying tail).

Known Limitations

• Single-sample VCF: *in trans* phase needs parental testing.
• Depends on HPO gene–phenotype coverage (partial for some MVA genes).
• Non-coding regulatory variants are excluded.
• Two Step 3 runs differ (905 vs 1,526 HIGH/MODERATE variants); the cause is open.

Compute: with reference data downloaded and the snpEff database built, Step 2 ≈ 1.5 min (10 cores; ≈ 7 min on 2), Step 3 minutes (bound by API response times), Steps 4–5 < 1 min. 2+ cores, 8 GB RAM, 8 GB disk; no cost beyond hardware and free APIs.